



Natural language processing course project report

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Abstract

We built an AI assistant for the UL FRI student office that answers administrative questions in Slovenian and English. The system is a RAG pipeline that crawls and parses UL/FRI web content, chunks it with a section-aware splitter, embeds the chunks with a multilingual sentence-transformer, and stores them in FAISS. At query time we compare three retrieval modes – dense, BM25+dense hybrid, and hybrid with a cross-encoder reranker – feeding an instruction-tuned generator. We evaluate the pipeline with both classical retrieval metrics ($\text{Recall}@k$, $\text{MRR}@k$, $\text{nDCG}@k$) and a RAGAS-style LLM-as-judge protocol scoring faithfulness, hallucination, refusal handling, and overall quality. Early on we noticed that many of our reference answers were not actually grounded in the corpus, so we first ran a Sonnet-graded audit and used the cleaned set for every comparison. An embedder $\times$ retrieval-mode sweep followed by a chunk-size sweep picks `multilingual-e5-base` with hybrid retrieval, a cross-encoder reranker, $\text{top-}k=4$, and 400-token chunks as the best configuration, with a Sonnet-graded in-scope faithfulness of 0.740 and 11/11 correct refusals on out-of-corpus questions.

Keywords

Retrieval augmented generation, dense retrieval, hybrid retrieval, multilingual embeddings, student affairs chatbot

Introduction

We are building an AI assistant for the student office of the Faculty of Computer and Information Science (UL FRI). The goal is a chatbot that answers everyday administrative questions about study programmes, regulations, and the office's work in Slovenian and English. Plain LLMs are a poor fit here: the answers depend on small details (deadlines, paragraph numbers, application forms) that need to come from the official documents, not from the model's training data. We therefore build the assistant on Retrieval Augmented Generation [1], which has been a good fit for university-style FAQ chatbots in prior work [2, 3] and which a recent survey shows helps with hallucination and out-of-date knowledge in educational deployments [4]. Our pipeline borrows the comparative-retrieval setup (dense vs. hybrid vs. reranked, with a chunk-size sweep) from RAG-for-admissions and university-resources studies [5, 6], and the choice to evaluate locally hosted instruction-tuned LLMs from work on open-day chatbots [7] and on RAG vs. fine-tuning for knowledge-intensive tasks [8].

Methods

Data collection and parsing

The pipeline is organised as *runs*: each run lives in its own directory holding the raw downloads, parsed

JSONL documents, chunks, and the FAISS index, so we can rebuild any single stage without touching the others. A lightweight crawler starts from a curated seed list and walks breadth-first, with separate depth limits for `fri.uni-lj.si`, the other `*.uni-lj.si` domains, and external domains. It downloads HTML pages along with their linked PDF/DOCX attachments, augments the crawl with sitemaps and RSS/Atom feeds, respects `robots.txt`, throttles requests, and stores SHA-256 hashes plus HTTP timestamps in a per-run `manifest.jsonl`; update crawls reuse `ETag/If-Modified-Since`. We then parse each document into a common JSON record (stable URL-derived `doc_id`, title, language, ordered named sections): HTML via BeautifulSoup (with optional `trafilatura`), PDF via Docling with Tesseract OCR or PyMuPDF as a fallback, DOCX via `python-docx`. Image-only sources are indexed via required sidebar text descriptions.

Chunking, embeddings and retrieval

We split documents with a section-aware sliding window measured in whitespace tokens (default 400, overlap 80). Every chunk is prefixed with the document title and section name, gets a stable `chunk_id`, and is then deduplicated with a SimHash filter. The chunks are embedded with `intfloat/multilingual-e5-base` – using the E5-

style passage:/query: prefixes and L2 normalisation – and indexed in FAISS as an `IndexFlatIP` (which is cosine under normalisation). At query time we support three retrieval modes: dense (top- k from FAISS), hybrid (BM25 fused with dense scores after min-max normalisation on the union of both top lists), and hybrid plus cross-encoder rerank (a sentence-transformers cross-encoder reranks a candidate pool of size 20). Retrieved chunks are then formatted into a numbered context block and a language-specific system prompt tells the generator to answer *only* from that block and in the detected language.

Evaluation set and reference audit

Our evaluation set is a hand-written JSONL file of 61 questions (51 *in-scope* – the corpus contains a factual answer – and 10 *negatives*, deliberate out-of-corpus probes used to test refusal). Each entry carries the expected language and a list of expected keywords; positives also carry `relevant_chunk_ids` for chunk-level retrieval metrics and a free-form `reference_answer` that we use as the human-written gold for the LLM-as-judge protocol below.

The first time we ran the judge, in-scope faithfulness was very low. Inspection showed many of our reference answers contained details the corpus actually does not support, so the judge was, correctly, marking them down. To stop this reference drift from contaminating later comparisons we run a Sonnet-graded fact-check before the cross-configuration sweeps: for each positive question we retrieve a top-30 BM25 $\cup$ dense pool from the current index and ask Claude Sonnet (`claude-sonnet-4-6`, $T=0$) to label every claim in the reference as `supported`, `partial`, or `unsupported`, and to optionally produce a grounded rewrite. The application policy is fixed: `supported` keeps the original; `partial` or `unsupported` with a rewrite uses the rewrite (flagged `auto_rewritten`); `unsupported` without a rewrite is dropped; negatives are never modified. The cleaned set is written as a versioned artefact `questions_full.v2.json`.

LLM-as-judge protocol

We use a RAGAS-style judge: for each row we send the question, the retrieved top- k chunks, the model answer, and the reference answer to Claude with a strict-JSON rubric covering *faithfulness*, *answer relevance*, and *context relevance* (each in $[0, 1]$), binary *hallucination* (per-answer flag, set when at least one claim is unsupported by the retrieved context) and *refusal* flags, and an integer *overall* score in $[1, 5]$. Negatives are scored under the same rubric, with a correct refusal rewarded with high faithfulness. All judge calls run at $T=0$ for reproducibility. To keep cost down without sacrificing the bottleneck quality we use Sonnet (`claude-sonnet-4-6`) for the reference audit and the final re-judgment, and Haiku (`claude-haiku-4-5`) for the comparison sweeps – the two judges agreed to within 0.001 in in-scope faithfulness on the winning configuration (0.739 vs. 0.740), so we trust Haiku for ranking. Per-configuration indices are built by short driver scripts that symlink `raw/` from the source crawl and pass

`EMBEDDING_MODEL/CHUNK_SIZE/CHUNK_OVERLAP` as environment overrides; the chosen parameters are persisted to `embedding_info.json` next to the index and re-checked on load, so we don't accidentally read an index with a different embedder or chunk size.

Results

Corpus, evaluation set, and reference audit

At the default 400/80 chunk policy the full crawl yields 2,742 parsed documents and 29,296 chunks, up from the 176 documents in the initial *Test* corpus we used during development. Building the index for the full crawl took roughly 4 hours when OCR was on; disabling OCR (PyMuPDF text extraction only) brought that down to ~ 14 minutes for the same ~ 500 PDFs and we did not see any measurable retrieval-quality drop afterwards, so we kept it off for all later experiments. Running the Sonnet audit on the original references against this corpus was a wake-up call: only 9/47 ($\sim 19\%$) of the in-scope references were fully supported as written. Of the rest, 31 were classified `partial` and rewritten using only chunk-attested facts, 7 were `unsupported` (3 dropped, 4 rewritten where Sonnet found alternative chunks), and 14 were skipped (no factual reference, or negative). The cleaned `questions_full.v2.json` contains 54 in-scope and 11 negatives and is the basis for everything that follows.

Chunk-level retrieval (Test corpus)

On the Test corpus, where we have chunk-level annotations, dense retrieval at $k=4$ already reaches 88.0% Recall@4 and 66.3% nDCG@4. Adding the cross-encoder reranker lifts these to 100.0% and 93.4%, while hybrid + rerank is essentially tied (100.0% / 93.9%). The corresponding chunk-size sweep on the same corpus showed sizes 200–500 all reaching $\sim 97.9\%$ Recall, with 300 giving the highest precision; the LLM-judged sweep on the full crawl, reported below, revisits this choice on a stricter metric and ends up picking 400.

Embedder $\times$ retrieval-mode sweep (LLM-as-judge)

Table 1 shows the embedder $\times$ retrieval-mode sweep at top- $k=4$ on the cleaned set, judged by Claude Haiku at $T=0$; refusal-side faithfulness is ≥ 0.90 across all six configurations. Two things stood out for us. First, the cross-encoder reranker is by far the biggest lever: on `e5-base` it lifts faithfulness from 0.575 (dense) and 0.565 (hybrid) to 0.739, and context relevance from about 0.72 to 0.857. Second, the result we did not expect – `e5-base + rerank` beats `e5-large + rerank` on every in-scope metric while embedding about $2.5\times$ faster. Once the cross-encoder does the final ranking, the smaller first-stage embedder is enough.

Chunk size on the winning configuration

Holding the winner fixed (`e5-base`, hybrid + rerank, $k=4$) and varying the chunk size with 20% overlap, the Haiku-judged in-scope faithfulness is 0.691 at 200/40 (50,546 chunks), 0.698 at 300/60 (36,331), and 0.739 at 400/80

Table 1. Embedder $\times$ retrieval-mode sweep on `questions_full_v2.json` (Haiku judge, top- $k=4$, $T=0$).

Embedder	Mode	Faithfulness	Hallucination	Ctx rel.	Overall (1–5)
multilingual-e5-base	dense	0.575	59.1%	0.725	3.000
multilingual-e5-base	hybrid	0.565	63.6%	0.715	2.841
multilingual-e5-base	hybrid + rerank	0.739	54.5%	0.857	3.432
multilingual-e5-large	dense	0.680	54.5%	0.812	3.250
multilingual-e5-large	hybrid	0.650	61.4%	0.787	3.205
multilingual-e5-large	hybrid + rerank	0.708	59.1%	0.866	3.364

(29,296). Larger chunks help here because the cross-encoder benefits from a bit more local context per chunk, and 400-token chunks also produce the smallest, fastest index of the three – so we keep them as the default.

Headline configuration and ablation

We then re-judged the winning configuration with Claude Sonnet at $T=0$ on the cleaned set (Table 2). The headline in-scope faithfulness is 0.740, answer relevance 0.820, context relevance 0.807, overall 3.605/5, and hallucination rate 44.2%; on the 11 negatives the system refuses correctly in all 11/11 cases with refusal-side faithfulness 0.905. Walking the gain through the stages: the original baseline (small corpus, noisy refs, dense $k=8$) sat at Haiku-judged faithfulness of 0.34; the full crawl plus reference audit at dense $k=4$ already moves it to 0.575 (+0.24); hybrid on its own does not help (0.565); adding the cross-encoder reranker takes it to 0.739 Haiku / 0.740 Sonnet (+0.16), with the per-answer hallucination rate dropping from 80.9% to 44.2%.

Table 2. Headline numbers for the winning configuration (Sonnet judge, $T=0$). The in-scope $n=43$ excludes 11 of the 54 cleaned questions where Haiku and Sonnet disagreed on the refusal flag during pre-screening; including them does not change the means by more than 0.005.

Metric	In-scope (n=43)	Refusal (n=11)
Faithfulness	0.740	0.905
Answer relevance	0.820	—
Context relevance	0.807	—
Overall (1–5)	3.605	3.818
Hallucination rate	44.2%	—
Refused correctly	—	11 / 11

Discussion

Both the retrieval-only and the LLM-as-judge results point at the same story. The biggest improvements split roughly evenly between two stages: corpus expansion and the Sonnet audit together move dense- $k=4$ faithfulness from 0.34 to 0.575 (about 58% of the total gain), and the cross-encoder reranker takes it from there to 0.740 (the remaining 42%). Hybrid retrieval on its own does almost nothing, which surprised us; it only starts to pay off once the reranker is on top of it, presumably be-

cause the union of BM25 and dense candidates gives the cross-encoder a more diverse pool to choose from. Another finding we did not expect was that `multilingual-e5-base` beats `multilingual-e5-large` once reranking is in play while embedding $\sim 2.5\times$ faster: the cross-encoder absorbs the work that would otherwise depend on a stronger first-stage embedder.

The biggest caveats are around the evaluation set: 54 in-scope and 11 negative questions are not many, the language is mostly Slovenian, and even with $T=0$ a single LLM judge has some per-question variability – a stratified manual review with Cohen’s κ against the Sonnet judge is the next thing on our list. The headline numbers are reported with a small Qwen generator that fits on a laptop; we have wired up an end-to-end comparison against `GaMS3-12B`, `Llama-3.1-8B`, and `Mistral-7B` on the cluster (same retrieval configuration, same judge), and a 3-question smoke test verified the flow, but the cluster jobs were not finished by the time of this submission. We also disabled OCR to keep full-crawl build time tractable: for scanned PDFs this would silently drop content, so a future iteration should re-run Docling/Tesseract just on the subset where PyMuPDF returns near-empty text.

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